Sahil Singh Diwan

Gen AI Engineer | Claude Certified Architect (Anthropic)
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Professional Summary

Gen AI engineer with 5+ years in software and AI engineering, the last two and a half **taking RAG pipelines, agentic workflows and LLM applications from proof of concept to production**. Built the retrieval engine behind an SRE incident-resolution assistant that grew from a 20-user POC to a 200+-SRE beta and cut incident lookup time ~70%. Co-led an agentic AI platform into alpha with a global semiconductor client and shipped Convogene.ai, a live RAG chatbot with near-zero hallucinations. Available to join immediately.

Technical Skills

- **Languages**: Python (primary), SQL
- **Generative AI**: RAG, agentic workflows, LangChain, LangGraph, MCP & agent skills, Gemini, Claude API, Azure OpenAI, prompt engineering, guardrails, Ragas, Langfuse / LangSmith
- **API Frameworks**: FastAPI, Flask, REST APIs, async Python (asyncio)
- **Libraries**: Hugging Face Transformers, PyTorch, TensorFlow, Keras, NumPy, Pandas, PyMuPDF, pdfplumber, PyPDF, Unstructured, Docling, BeautifulSoup, Scrapy, Playwright
- **Distributed Systems**: Apache Kafka (event streaming and fan-out), Apache Airflow, event-driven microservices
- **DevOps**: Docker, Kubernetes, Helm, Terraform (infrastructure as code), GitHub Actions CI/CD, Git, Allure reporting
- **Vector Databases**: FAISS, Milvus, ChromaDB, Elasticsearch
- **Databases**: PostgreSQL, MongoDB, MySQL
- **Cloud**:

GCP	Vertex AI (Gemini), Google Kubernetes Engine (GKE), Cloud Storage, IAM & service accounts, Secret Manager
Azure	Azure OpenAI, Azure Kubernetes Service (AKS), Container Apps, Blob Storage, Key Vault
AWS	EC2, S3, EKS, Bedrock, Secrets Manager, Elastic Load Balancing
IBM Cloud	Code Engine, Container Registry, Kubernetes Service (IKS), watsonx.ai, Cloud Object Storage, IAM, Secrets Manager

Professional Experience

Infobell IT Solutions — **Bengaluru, India**
Software Engineer, AI — *Mar 2024 - Oct 2026*

Roles and responsibilities

- Co-led an agentic AI platform with one peer and mentored the junior engineers building it.
- Owned the design and delivery of execution layers end to end: architecture, CI/CD, credentials and role-based access control, and reporting.
- Took GenAI systems from proof of concept to production for enterprise clients and handed them over to the clients' in-house teams.
- Worked directly with client stakeholders: demoed the platform's alpha to project team leads and measured results with the SREs using the system.
- Ran internal enablement sessions on Claude Code and Cursor for the engineering team.

Projects

- **Agentic AI Test-Automation Platform (global semiconductor client)**
  Tech stack: Python, Gemini on GCP Vertex AI, on-premises LLMs, Azure OpenAI, GitHub Actions, Playwright, Allure
  Clones a repository, maps its call graph with LLMs, then generates and self-corrects API, workflow and security test suites, with pull-request blast-radius analysis. Owned test execution: an LLM self-healing loop that repairs failing tests mid-run, the multi-provider model layer, credentials and role-based access control.

- **SRE Incident-Resolution Assistant (major public cloud)**
  Tech stack: Python, Apache Kafka, Elasticsearch, hybrid retrieval, Kubernetes, Helm, ServiceNow, GitHub
  Built the retrieval engine and Kafka event pipeline that scaled from a 20-user POC to a 200+-SRE beta and cut time-to-find similar incidents ~70% (15-20 min to ~5 min). Rebuilt failing semantic search as hybrid retrieval with structured filtering and identifier masking, taking recall from ~30% to ~80%.
- **Convogene.ai, production RAG chatbot (semiconductor client)**
  Tech stack: Python, LangGraph, LangChain, Gemini, RAG, BeautifulSoup, Scrapy, Microsoft Azure
  Built and shipped a live chatbot, convogene.ai, that answers detailed processor-lineup questions from scheduled scraping and ingestion. Two-layer data and prompt guardrails took hallucinations from frequent to near-zero.
- **Cloud-Onboarding Assistant (Fortune-50 cloud provider)**
  Tech stack: Python, agent skills, Terraform, cloud IAM
  Designed the execution layer of an assistant that moves users to a new cloud platform inside a two-week trial. Its agent skill authenticates to the user's account, manages resources and provisions infrastructure via Terraform.
- **Linux Kernel-Backporting Automation**
  Tech stack: Python, Git, LLM agent
  Built tooling that auto-applies ordered sequences of 5-120 kernel patches, work that used to take engineers months by hand. It localizes each failure for engineer review instead of stopping the run.
- **Cloud-Native Anomaly Detection**
  Tech stack: Python, Apache Airflow, Kubernetes, Pandas, NumPy
  Built and deployed an automated anomaly-detection system processing data-center-scale datasets on Kubernetes.
- **Generative Media (multimodal)**
  Tech stack: PyTorch, Hugging Face, instruct-pix2pix
  Fine-tuned an instruct-pix2pix image-editing model for a generative-media solution. Owned the application rollout.

India STEM Foundation — **Remote & On-site, India**
Master Trainer, AI & Python — *Aug 2022 - Aug 2023*
- Led online Python and AI training for students in the USA, UK, Singapore and India, and designed hands-on robotics curricula (Python, C, Raspberry Pi, Arduino).

Koderoom — **Remote, India**
Junior Software Developer — *Jun 2020 - May 2021, Jul 2021 - Jun 2022*
- Built and maintained backend APIs for a legal contract-management platform: document parsing, data structuring, and secure retrieval of sensitive contracts.

Other Experience (Non-IT)

Think and Learn Pvt. Ltd. (BYJU'S) — **India**
Business Development Trainee (Sales) — *Jun 2021 - Jul 2021*
- One-month sales traineeship (01 Jun 2021 - 08 Jul 2021), taken between two periods at Koderoom.

Shubham Enterprises, deployed at Samiya Engineering Works — **Nagpur, India**
Site Engineer — *Aug 2018 - Jun 2019*
- On-site mechanical engineering role, prior to moving into software.

Education

Centre for Development of Advanced Computing (C-DAC) — **India**
Post Graduate Diploma (6 months) — *Sep 2023 - Feb 2024*

Rashtrasant Tukadoji Maharaj Nagpur University, Oberoi Centre for Excellence — **Nagpur, India**
Post Graduate Diploma, Industrial Robotics — *2019 - 2020*

G H Raisoni Academy of Engineering and Technology — **Nagpur, India**
B.E., Mechanical Engineering — *2015 - 2018*

Certifications

- **Claude Certified Architect, Foundations** (Anthropic): issued Aug 2026, valid to Aug 2027. Credly badge.
- **IBM Cloud Advocate Essentials**: completed Dec 2025.

Languages

- English (professional), Hindi (native), Marathi (native).